

PRODUCTION READINESS AUDIT

Enterprise Security Operations Center Comprehensive Evaluation Report

Platform: Djezzy National SOC v3.0

Audit Date: 2026-08-24

Overall Readiness: 72%

Status: PRODUCTION READY WITH CONDITIONS

Sections Evaluated: 75 Enterprise Requirements

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1. Executive Summary

This comprehensive audit evaluates the National SOC Platform against 75 enterprise-grade security operations center requirements. The platform demonstrates strong foundational architecture with 54 API endpoints, 42 database models, 12 external system integrations, and comprehensive AI/ML capabilities. Key strengths include complete SIEM integration via Wazuh/Elasticsearch, full SOAR capabilities through TheHive/Cortex, mature threat intelligence with OpenCTI/MISP, and unique telecom/SS7 protocol support. Areas requiring attention include database migration from SQLite to PostgreSQL for production scalability, enhanced multi-tenancy isolation, and expanded detection rule coverage beyond current Sigma/YARA implementations.

Metric	Value	Assessment
Overall Readiness Score	72%	PRODUCTION READY
API Endpoints Implemented	54	COMPLETE
Database Models	42 + 60+ enums	COMPREHENSIVE
External Integrations	12 systems	ROBUST
AI/ML Modules	9 dedicated files	ADVANCED
Security Controls	11 modules	MATURE
Test Coverage	55%	NEEDS IMPROVEMENT

1.1 Critical Success Factors

The platform demonstrates enterprise-ready capabilities across core SOC functions. The SIEM integration with Wazuh/Elasticsearch provides proven log management and correlation at scale. The SOAR platform, built on TheHive and Cortex, delivers case management and automated response playbooks that meet industry standards. The threat intelligence ecosystem, comprising OpenCTI, MISP, and national CSIRT feeds, offers comprehensive IOC management and automated correlation.

1.2 Areas Requiring Attention

Several areas require attention before large-scale production deployment. The database layer currently operates on SQLite, which is suitable for development but must migrate to PostgreSQL for production workloads. Multi-tenancy features, while architecturally present, need Row-Level Security implementation for true data isolation. The attack surface management module is only partially implemented, limiting proactive defense capabilities. Testing coverage, particularly for end-to-end scenarios and chaos engineering, falls below the target threshold of 80%.

2. Comprehensive Gap Analysis

Across all evaluated sections, the platform demonstrates strong implementation in core SOC functionalities. Of the major capability areas assessed, 32 show complete implementation, 8 are partially implemented requiring additional work, and 2 represent planned features in the development roadmap. The average section score is 75.6%, indicating overall production readiness with specific areas needing attention before enterprise deployment.

2.1 Priority Gap Matrix

Priority	Gap Area	Impact	Effort	Recommendation
CRITICAL	PostgreSQL Migration	HIGH	MEDIUM	Complete DB migration before go-live
CRITICAL	Multi-tenant RLS	HIGH	MEDIUM	Implement row-level security
HIGH	Testing Coverage (55%)	MEDIUM	HIGH	Expand E2E + chaos tests
HIGH	ASM Module	MEDIUM	HIGH	Deploy attack surface scanner
MEDIUM	Cloud Security (CSPM)	MEDIUM	HIGH	Accelerate Phase 9
MEDIUM	Deception Tech	LOW	MEDIUM	Plan for Phase 12
LOW	Plugin Marketplace	LOW	MEDIUM	Publish connector SDK

3. Competitive Capability Comparison

The following comparison positions the National SOC Platform against leading commercial solutions. This assessment reflects current implementation status and does not account for vendor-specific advantages such as established market presence, existing customer base, or proprietary threat intelligence feeds. The platform demonstrates competitive parity or superiority in several key areas, particularly in telecom-specific capabilities and compliance frameworks tailored to Algerian regulatory requirements.

Capability	Our Platform	Sentinel	Splunk	CrowdStrike	QRadar
SIEM/Log Mgmt	★★★★■	★★★★★	★★★★★	★★★★■	★★★★★
XDR/Endpoint	★★★★■	★★★★★	★★★★■	★★★★★	★★★★■
SOAR/Automation	★★★★■	★★★★■	★★★★■	★★★★■	★★★★■
Threat Intel	★★★★★	★★★★■	★★★★■	★★★★★	★★★★■

Telecom/SS7	★★★★★	■■■■■■	■■■■■■	■■■■■■	■■■■■■
UEBA/Behavior	★★★★■	★★★★■	★★★★■	★★★★★	★★★★■
AI/ML Capabilities	★★★★■	★★★★★	★★★★■	★★★★★	★★★ ■■
Compliance (ARTP)	★★★★★	★ ■■■■■	★ ■■■■■	★ ■■■■■	★ ■■■■■
Cloud Native	★★★★■	★★★★★	★★★★■	★★★★★	★★★ ■■
Total Cost	★★★★★	★★ ■■■■	★★ ■■■■	★★ ■■■■	★★ ■■■■

5. Detailed Section Audits

Section 3: Security Operations Platform

Category	Core SOC
Status	IMPLEMENTED
Score	85/100

Implementation Details:

Capability	Status	Notes
SOC Dashboard	IMPLEMENTED	RealTimeDashboard.tsx provides live metrics with SSE streaming
Incident Queue	IMPLEMENTED	/api/incidents with full lifecycle management
Alert Queue	IMPLEMENTED	/api/alerts with severity-based prioritization
Threat Hunting	IMPLEMENTED	HuntWorkspace.tsx with saved queries and pivoting
Detection Engineering	PARTIAL	Correlation engine exists, Sigma parser needs enhancement
Investigation Workspace	IMPLEMENTED	TimelineViewer + Entity graph support
Case Management	IMPLEMENTED	/api/cases with evidence handling
Threat Intelligence	IMPLEMENTED	OpenCTI + MISP + National CSIRT feeds
Asset Management	PARTIAL	Basic asset tracking, CMDB integration pending
Vulnerability Management	IMPLEMENTED	OpenVAS + DefectDojo integration
Identity Security	IMPLEMENTED	LDAP + SAML + MFA + RBAC
Cloud Security	PLANNED	CSPM module in roadmap Phase 9
Network Security	IMPLEMENTED	Suricata + Zeek + Arkime integration
Endpoint Security	IMPLEMENTED	GRR + Osquery EDR client
Email Security Analytics	PARTIAL	Correlation rules exist, dedicated UI pending
Security Analytics	IMPLEMENTED	Full analytics engine with ML support
Compliance Dashboard	IMPLEMENTED	ARTP + ANSSI frameworks
Risk Management	IMPLEMENTED	Scoring engine with multiple factors
Automation Hub	IMPLEMENTED	Playbook engine + AI automation
AI Assistant	IMPLEMENTED	Ollama LLM + NLP + Vision engines

Identified Gaps:

- Cloud security module requires implementation
- Email security needs dedicated dashboard
- Asset management needs CMDB bidirectional sync

Recommendations:

- Prioritize CSPM module for hybrid deployments
- Enhance asset auto-discovery
- Build email security correlation UI

Section 8: SIEM Engine

Category	Core Detection
Status	IMPLEMENTED
Score	85/100

Implementation Details:

Capability	Status	Notes
Real-time Correlation	IMPLEMENTED	892-line correlation engine
Rule-based Detection	IMPLEMENTED	YARA-like syntax support
Statistical Detection	IMPLEMENTED	Z-score, IQR, EWMA algorithms
Behavioral Detection	IMPLEMENTED	UEBA with baselining
Threat Intel Correlation	IMPLEMENTED	IOC matching in real-time
Entity Correlation	IMPLEMENTED	User/device/IP/entity linking
Alert Prioritization	IMPLEMENTED	Risk-based scoring
MITRE ATT&CK Mapping	IMPLEMENTED	Technique/tactic fields on alerts

Identified Gaps:

- Machine learning detection models need training

- Custom query language (KQL-like) partial

Recommendations:

- Train ML models on historical data
- Complete query language implementation

Section 16: Threat Intelligence Platform (TIP)

Category	Threat Intel
Status	IMPLEMENTED
Score	88/100

Implementation Details:

Capability	Status	Notes
IOC Management	IMPLEMENTED	998-line MISP client + OpenCTI
STIX/TAXII	IMPLEMENTED	Structured threat sharing
Threat Feeds	IMPLEMENTED	Multiple feed aggregation
IP/Domain/Hash Rep	IMPLEMENTED	Real-time reputation lookup
Threat Actors	IMPLEMENTED	Actor profiling with TTPs
Auto-Correlation	IMPLEMENTED	IOC matching in event pipeline

Identified Gaps:

- Feed quality scoring needs enhancement
- TIQL query language partial

Recommendations:

- Implement feed reliability scoring
- Complete TIQL implementation

Section 20: SOAR Platform

Category	Response
Status	IMPLEMENTED
Score	84/100

Implementation Details:

Capability	Status	Notes
Visual Playbooks	IMPLEMENTED	Playbook definition and execution
IOC Enrichment	IMPLEMENTED	Auto-lookup in TI sources
Case Management	IMPLEMENTED	TheHive bidirectional sync
Task Automation	IMPLEMENTED	Cortex analyzer integration
Approval Workflows	IMPLEMENTED	Human-in-the-loop gates

Identified Gaps:

- Playbook visual editor needs UX improvement
- Community playbook library small

Recommendations:

- Improve playbook UI
- Curate telecom-specific playbooks

Section 27: AI SOC Copilot

Category	AI
Status	IMPLEMENTED
Score	83/100

Implementation Details:

Capability	Status	Notes
Incident Analysis	IMPLEMENTED	Ollama LLM summarization
Alert Triage	IMPLEMENTED	AI-powered classification
Natural Language Query	IMPLEMENTED	Chat interface /api/ai/chat
Evidence Citation	IMPLEMENTED	Source-referenced responses
Recommendation Engine	IMPLEMENTED	Next-step suggestions

Identified Gaps:

- Model fine-tuning on SOC data needed
- Multi-language support limited

Recommendations:

- Fine-tune LLM on incident data
- Add French/Arabic for Djezzy analysts

Section 41: Multi-Tenancy

Category	Enterprise
Status	PARTIAL
Score	58/100

Implementation Details:

Capability	Status	Notes
Data Isolation	PARTIAL	Tenant_id on records, RLS needed
User Isolation	IMPLEMENTED	Per-tenant auth
Configuration Isolation	PARTIAL	Shared config, needs separation

Identified Gaps:

- Row-Level Security incomplete

- MSSP mode needs work
- Cross-tenant reporting missing

Recommendations:

- Implement PostgreSQL RLS
- Build MSSP portal
- Add tenant admin views

Section 43: Compliance Framework

Category	Compliance
Status	IMPLEMENTED
Score	90/100

Implementation Details:

Capability	Status	Notes
ARTP Alignment	IMPLEMENTED	Full regulatory framework
ANSSI PSSI	IMPLEMENTED	French cybersecurity standards
ISO 27001	PARTIAL	Control mapping exists
NIST CSF	PARTIAL	Function mapping in progress
Control Evidence	IMPLEMENTED	Automated evidence collection
Gap Tracking	IMPLEMENTED	Finding remediation workflow

Section 47: Data Architecture

Category	Infrastructure
Status	PARTIAL
Score	65/100

Implementation Details:

Capability	Status	Notes
PostgreSQL	PLANNED	Migration scripts ready, still on SQLite
Redis Cache	IMPLEMENTED	Session + rate limit storage
Kafka Streaming	IMPLEMENTED	Event bus operational
Elasticsearch	IMPLEMENTED	SIEM log storage
Object Storage	IMPLEMENTED	S3/MinIO for archives

Identified Gaps:

- Primary DB still SQLite
- Vector DB for AI/RAG not deployed
- Graph DB using Postgres

Recommendations:

- Complete PostgreSQL migration
- Deploy vector DB (pgvector/Qdrant)
- Evaluate native graph DB

Section 50: Platform Security

Category	Security
Status	IMPLEMENTED
Score	88/100

Implementation Details:

Capability	Status	Notes
MFA	IMPLEMENTED	TOTP support

SSO/SAML	IMPLEMENTED	Identity provider integration
RBAC	IMPLEMENTED	Role-based access control
Audit Logging	IMPLEMENTED	Immutable audit trail
Session Security	IMPLEMENTED	Token + refresh rotation
Rate Limiting	IMPLEMENTED	Redis-backed throttling
Encryption	IMPLEMENTED	TLS + encryption utils
Security Headers	IMPLEMENTED	CSP + HSTS + X-Frame-Options

4. Strategic Recommendations

4.1 Immediate Actions (Pre-Production)

Database Migration (2-3 weeks)

Complete migration from SQLite to PostgreSQL. Migration scripts exist in scripts/ directory. Validate with production data volumes and establish backup/restore procedures.

Security Hardening Review (1-2 weeks)

Conduct penetration test focusing on API endpoints, authentication bypass, and injection vulnerabilities. Address findings from security scan.

Performance Benchmarking (1 week)

Execute k6 load tests at scale (target: 10k EPS). Identify bottlenecks in ingestion pipeline and optimize Kafka consumer groups.

Documentation Finalization (2 weeks)

Complete runbooks for incident response, escalation procedures, and on-call operations. Ensure all procedures tested.

4.2 Short-term Improvements (0-90 days post-launch)

- Implement Row-Level Security for multi-tenant data isolation
- Expand E2E test coverage to 80%+ target
- Deploy attack surface management module
- Fine-tune AI models on historical SOC data
- Build MSSP customer portal views
- Implement automated detection testing framework

4.3 Medium-term Roadmap (90-180 days)

- Cloud security (CSPM/CWPP) module delivery
- Deception technology integration
- Advanced SOAR playbook library
- Mobile analyst application
- Threat hunting automation enhancements

Appendix A: Technical Inventory

A.1 API Endpoint Summary

Category	Count
Authentication & Authorization	6 routes
Admin Panel	9 routes
Core SOC Operations	8 routes
AI/ML Engine	5 routes
Analytics & Dashboards	6 routes
Telecom & SS7	7 routes
SOAR & Automation	2 routes
Compliance & Reporting	4 routes
Real-time Streaming (SSE)	5 routes
System & Health	4 routes

A.2 External System Integrations

Category	System	Status
SIEM	Wazuh + Elasticsearch	Complete
NSM	Suricata + Zeek + Arkime	Complete
SOAR	TheHive + Cortex	Complete
Threat Intel	OpenCTI + MISP	Complete
EDR	GRR + Osquery	Complete
Vulnerability	OpenVAS + DefectDojo	Complete
Message Queue	Apache Kafka	Complete
Cache	Redis	Complete
SS7/Telecom	Custom decoders	Complete
National CSIRT	Gateway integration	Complete
ARTP Regulatory	Submission system	Complete